

Report of Entropy estimates based on NIST SP 800-90B non-IID track

2026-May-09 12:34:24.126992

1 Identification information

1.1 Identification of acquisition data from entropy source

Table 1 Identification information of acquisition data from entropy source

|                                                  |                                                                                            |
|--------------------------------------------------|--------------------------------------------------------------------------------------------|
| URL of the acquisition data                      | https://github.com/usnistgov/SP800-90B_EntropyAssessment/blob/master/bin/truerand_8bit.bin |
| SHA-256 hash value of the acquisition data [hex] | c7e56911 d2657fa9 b6e86c03 d4477474 d6ec6986 91c5f32d 3918ec51 3713e3c3                    |
| Last write time                                  | 2021-Sep-20 10:22:46                                                                       |

- Name of the submitter of the acquisition data :
- Brief explanation of the acquisition data (or entropy source) :

1.2 Identification of analysis environment

Table 2 Identification information of analysis environment

|                      |                        |                                                        |
|----------------------|------------------------|--------------------------------------------------------|
| Analysis tool        | Name                   | Another entropy estimation tool with extensions        |
|                      | Versioning information | 1.1.7                                                  |
|                      | built as               | 64-bit application                                     |
|                      | built by               | Intel C++ Compiler ( __INTEL_LLVM_COMPILER: 20250303 ) |
|                      | linked libraries       | Boost C++ 1.91.0                                       |
| Analysis environment | Hostname               | ██████████                                             |
|                      | CPU information        | Intel(R) Core(TM) i5-██████████                        |
|                      | Physical memory size   | ██████ MiB                                             |
|                      | OS name                | Microsoft Windows 11 Pro                               |
|                      | OS version             | 10.0.26200 N/A Build 26200                             |
|                      | System type            | 64-bit                                                 |
|                      | Username               | ██████                                                 |

1.3 Identification of analysis conditions

Table 3 Identification information of analysis conditions

|                                                             |                                  |
|-------------------------------------------------------------|----------------------------------|
| Number of samples                                           | 1000000                          |
| Bits per sample                                             | 8                                |
| Byte to bit conversion                                      | Most Significant bit (MSb) first |
| Use Longest Common Prefix* <sup>1</sup> for 6.3.5 and 6.3.6 | True                             |

1.4 Identification of analysis method

NIST SP 800-90B [1] 6.3 with corrections [2] is applied

<sup>\*1</sup> See [3] and [4]

2

Executive summary

2.1

Numerical results of min-entropy estimates based on non-IID track

Table 4 Numerical results

| Estimator                                                                                                                     | $H_{\text{original}}^{\text{a}}$<br>[bit / 8 - bit] | Notes to $H_{\text{original}}$ | $H_{\text{bitstring}}^{\text{b}}$<br>[bit / 1 - bit] | Notes to $H_{\text{bitstring}}$ |
|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|--------------------------------|------------------------------------------------------|---------------------------------|
| The Most Common Value Estimate                                                                                                | 7.86512                                             | see 3.1                        | 0.998199                                             | see 4.1                         |
| The Collision Estimate                                                                                                        | —                                                   | —                              | 0.95841                                              | see 4.2                         |
| The Markov Estimate                                                                                                           | —                                                   | —                              | 0.999439                                             | see 4.3                         |
| The Compression Estimate                                                                                                      | —                                                   | —                              | 0.904233                                             | see 4.4                         |
| The t-Tuple Estimate                                                                                                          | 7.86512                                             | see 3.2                        | 0.933569                                             | see 4.5                         |
| The Longest Repeated Substring (LRS) Estimate                                                                                 | 7.9392                                              | see 3.3                        | 0.998671                                             | see 4.6                         |
| Multi Most Common in Window Prediction Estimate                                                                               | 7.98858                                             | see 3.4                        | 0.999563                                             | see 4.7                         |
| The Lag Prediction Estimate                                                                                                   | 7.93976                                             | see 3.5                        | 0.998402                                             | see 4.8                         |
| The MultiMMC Prediction Estimate                                                                                              | 7.92681                                             | see 3.6                        | 0.99966                                              | see 4.9                         |
| The LZ78Y Prediction Estimate                                                                                                 | 7.91928                                             | see 3.7                        | 0.998465                                             | see 4.10                        |
| The initial entropy source estimate [bit / 8 - bit]<br>$H_I = \min(H_{\text{original}}, 8 \times H_{\text{bitstring}})$       | 7.23386                                             |                                |                                                      |                                 |
| <sup>a</sup> Entropy estimate of the sequential dataset [source: NIST SP 800-90B [1] 3.1.3]                                   |                                                     |                                |                                                      |                                 |
| <sup>b</sup> An additional entropy estimation (per bit) for the non-binary sequential dataset [see NIST SP 800-90B [1] 3.1.3] |                                                     |                                |                                                      |                                 |

## 2.2 Visual comparison of min-entropy estimates from original samples

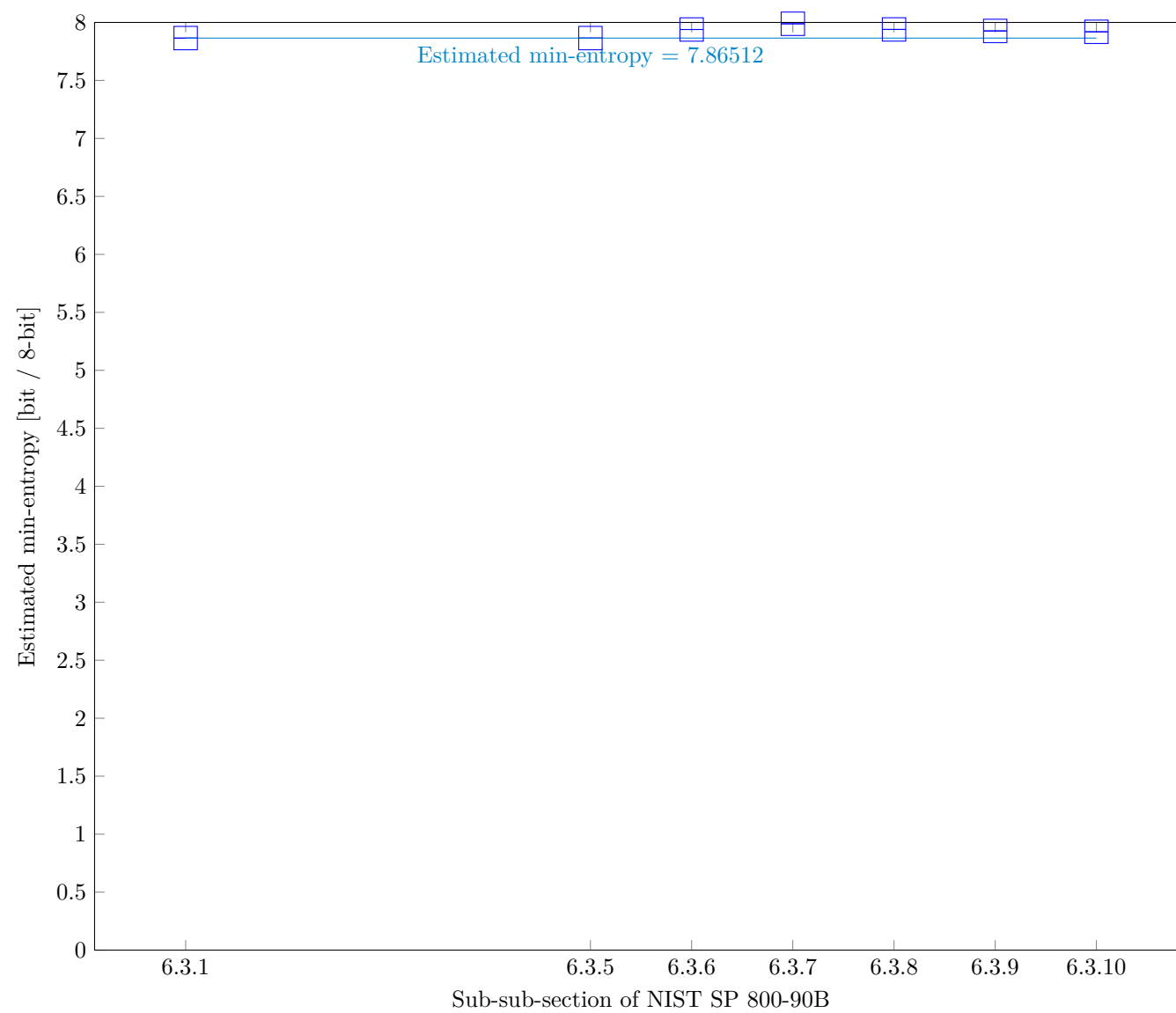


Fig. 1 Estimated Min-Entropy using §6.3 of NIST SP 800-90B

## 2.3 Visual comparison of min-entropy estimates by interpreting each sample as bitstring

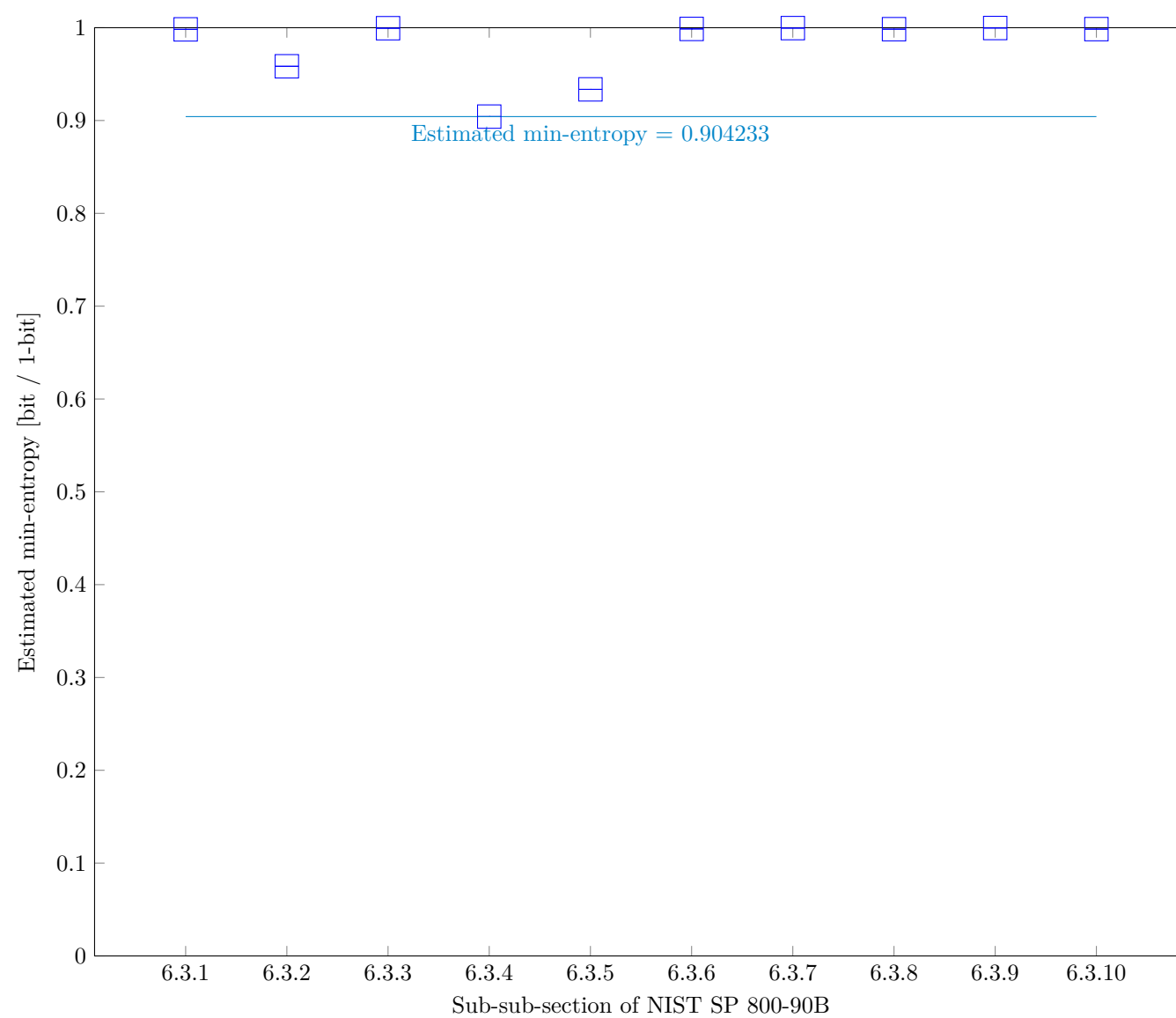


Fig. 2 Estimated Min-Entropy using §6.3 of NIST SP 800-90B

### 3 Detailed results of analysis from original samples

#### 3.1 The Most Common Value Estimate (NIST SP 800-90B Section 6.3.1)

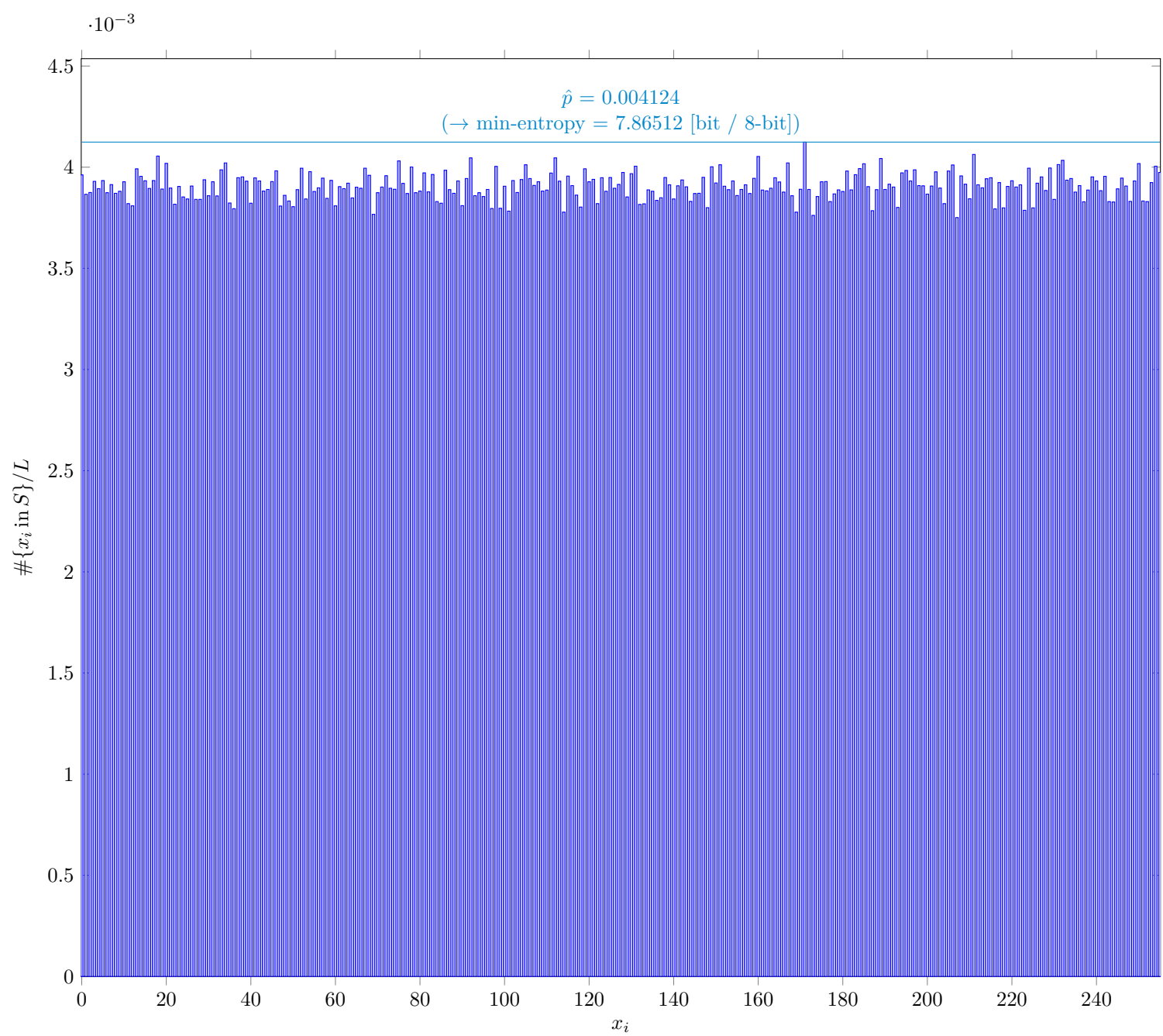


Fig. 3 Distribution of  $x_i$

##### 3.1.1 Supplemental information for traceability

Table 5 Supplemental information for traceability (NIST SP 800-90B Section 6.3.1)

| Symbol    | Value      |
|-----------|------------|
| mode      | 4124       |
| $\hat{p}$ | 0.004124   |
| $p_u$     | 0.00428907 |

### 3.2 The t-tuple Estimate (NIST SP 800-90B Section 6.3.5)

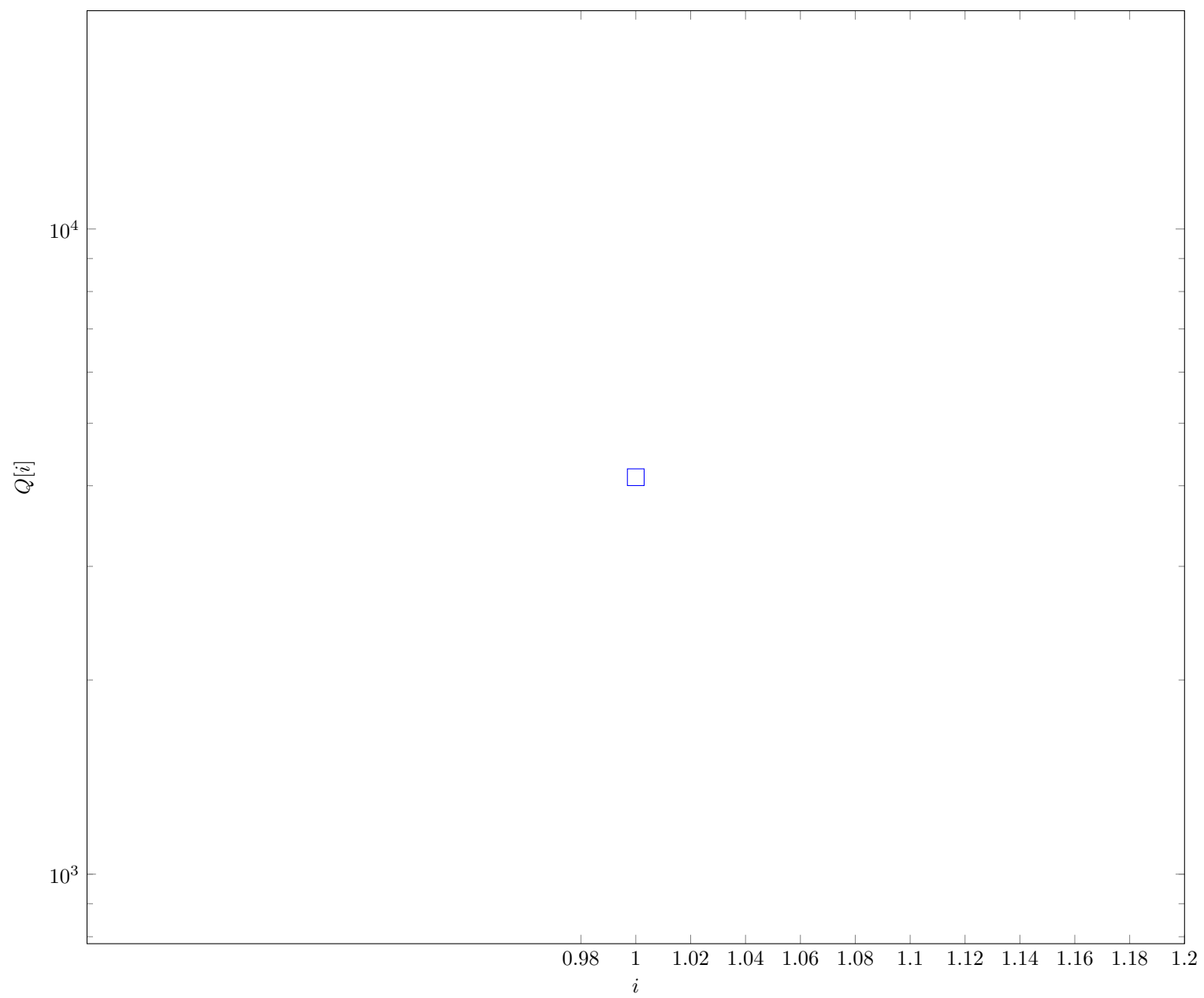


Fig. 4 Intermediate value  $Q[i]$  in §6.3.5 of NIST SP 800-90B



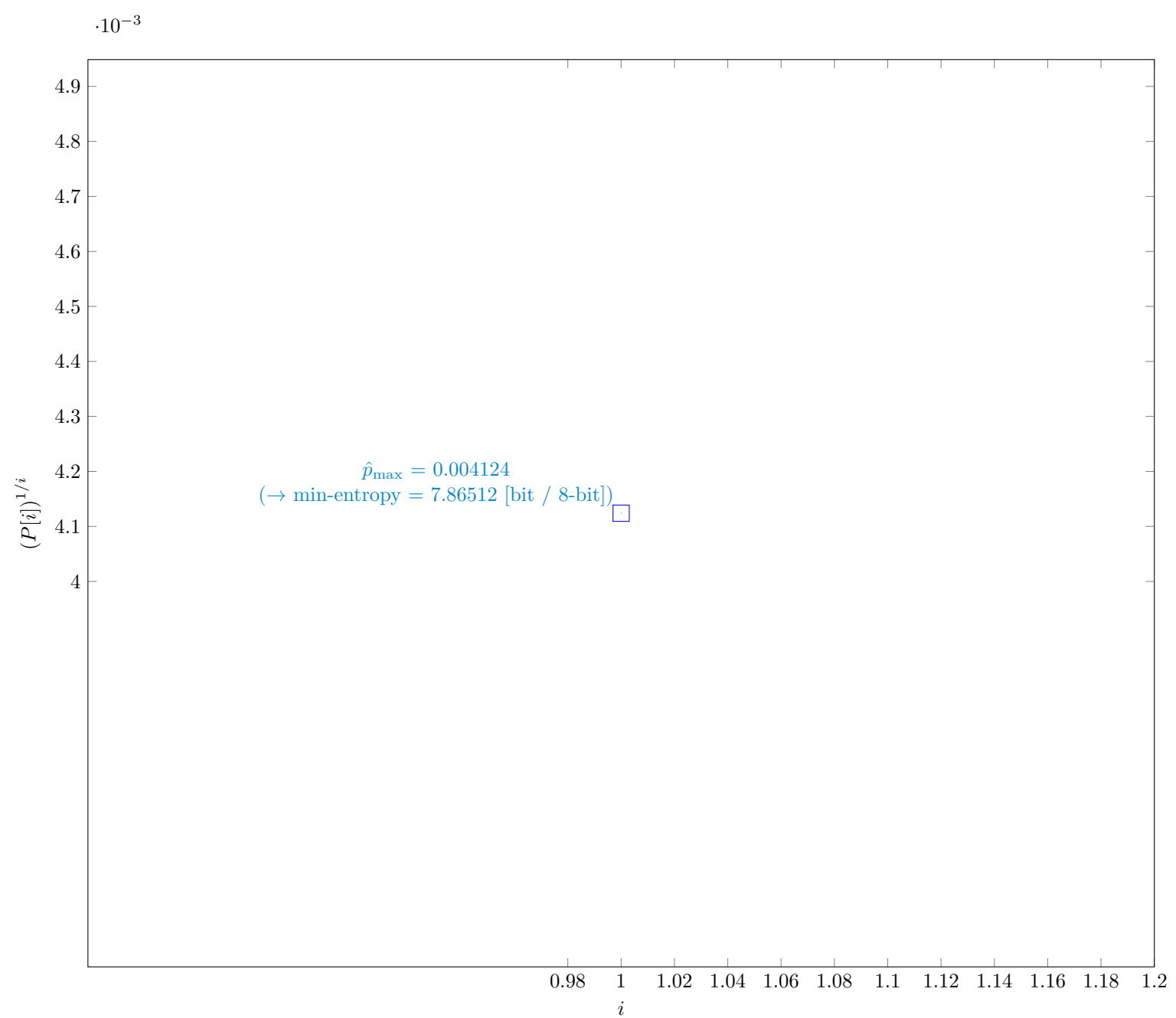


Fig. 5  $P[i]^{1/i}$  in §6.3.5 of NIST SP 800-90B

3.2.1 Supplemental information for traceability

Table 6 Supplemental information for traceability (NIST SP 800-90B Section 6.3.5)

| Symbol           | Value      |
|------------------|------------|
| $t$              | 1          |
| $\hat{p}_{\max}$ | 0.004124   |
| $p_u$            | 0.00428907 |



### 3.3 The LRS Estimate (NIST SP 800-90B Section 6.3.6)

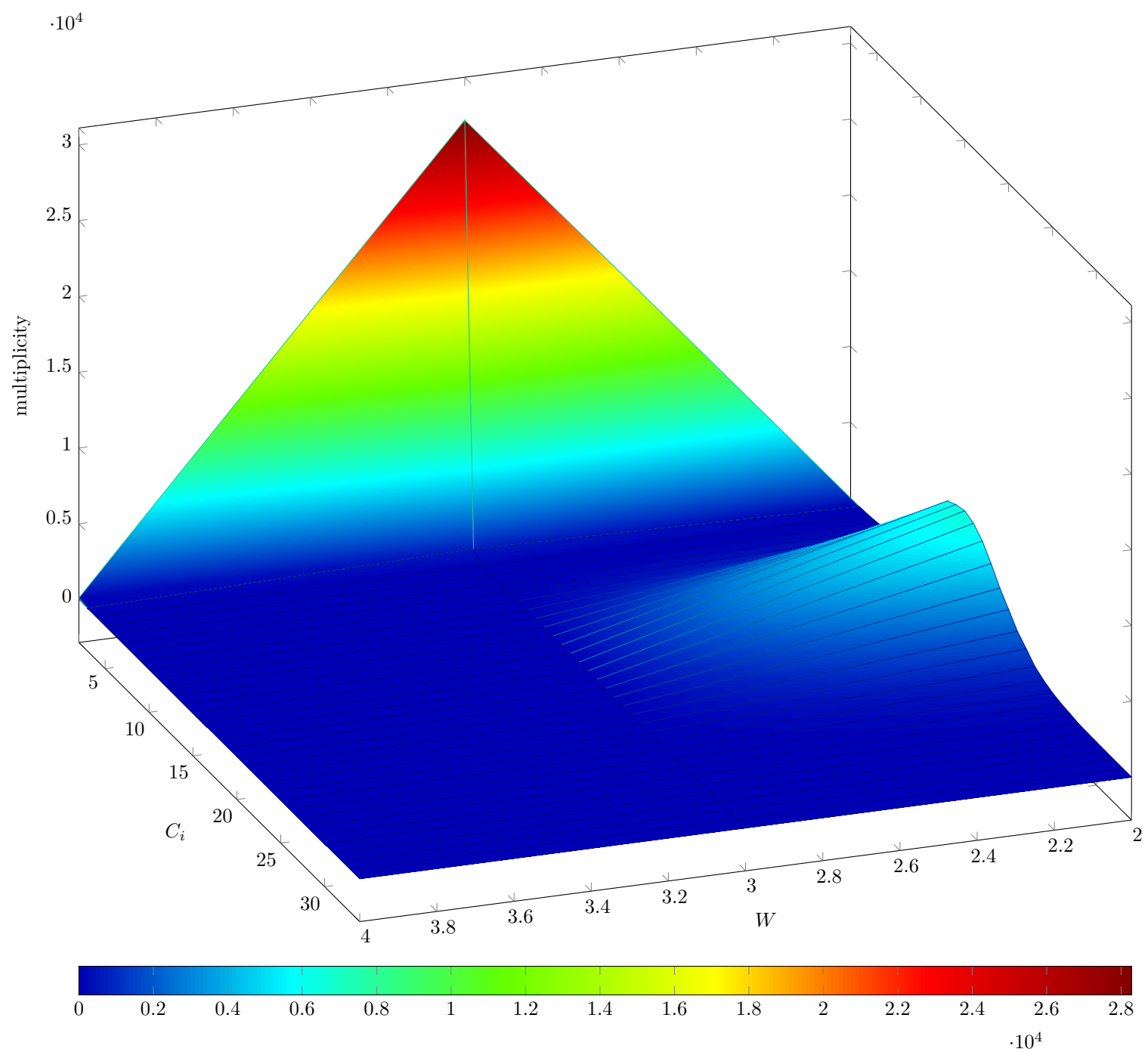


Fig. 6 Distribution of the multiplicity of  $W$ -tuple having the same  $C_i$  value in Step 3 of §6.3.6 of NIST SP 800-90B

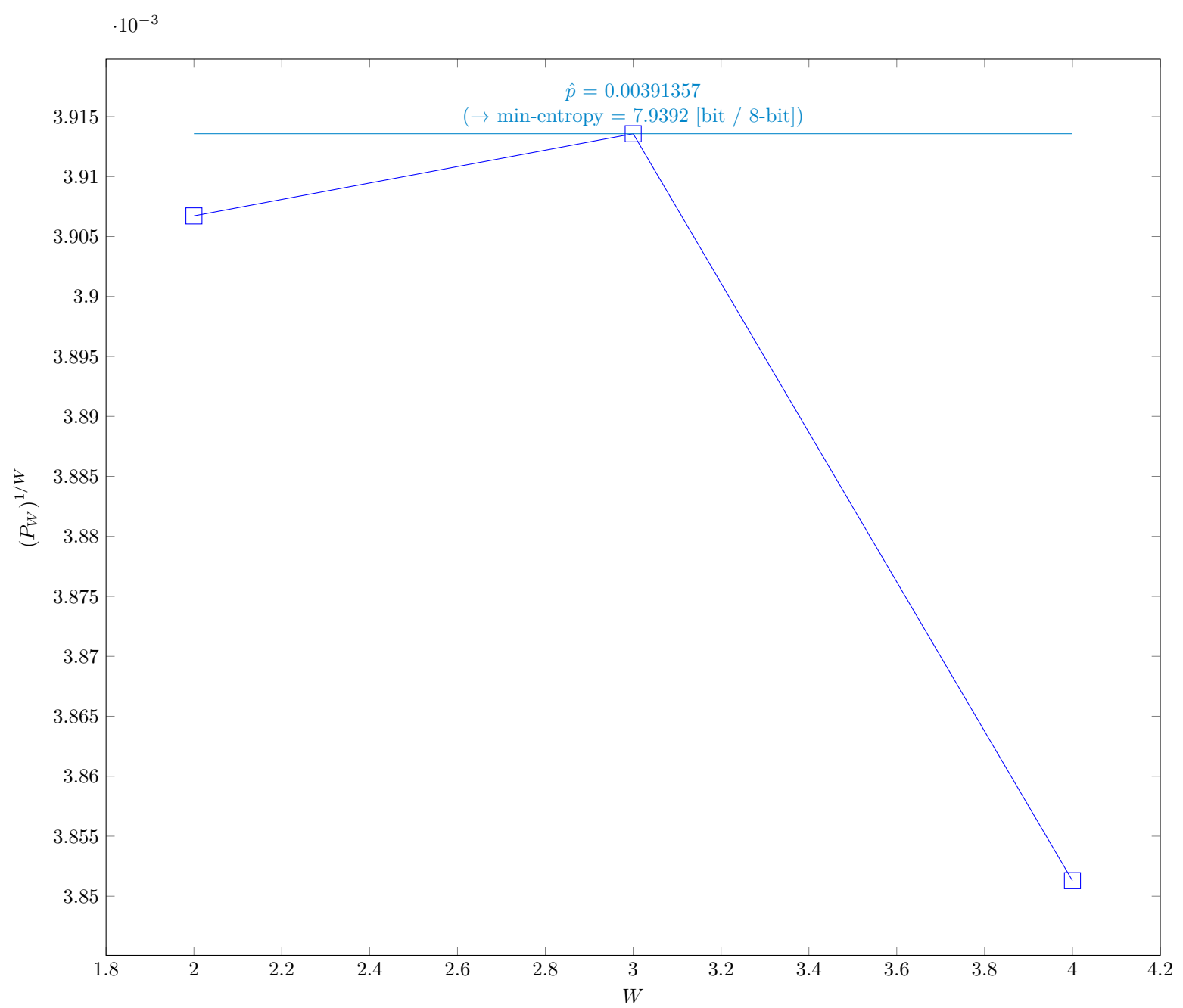


Fig. 7 Estimated average collision probability per string symbol in Step 3 of §6.3.6 of NIST SP 800-90B

3.3.1 Supplemental information for traceability

Table 7 Supplemental information for traceability (NIST SP 800-90B Section 6.3.6)

| Symbol    | Value      |
|-----------|------------|
| $u$       | 2          |
| $v$       | 4          |
| $\hat{p}$ | 0.00391357 |
| $p_u$     | 0.00407439 |

3.4 Multi Most Common in Window Prediction Estimate (NIST SP 800-90B Section 6.3.7)

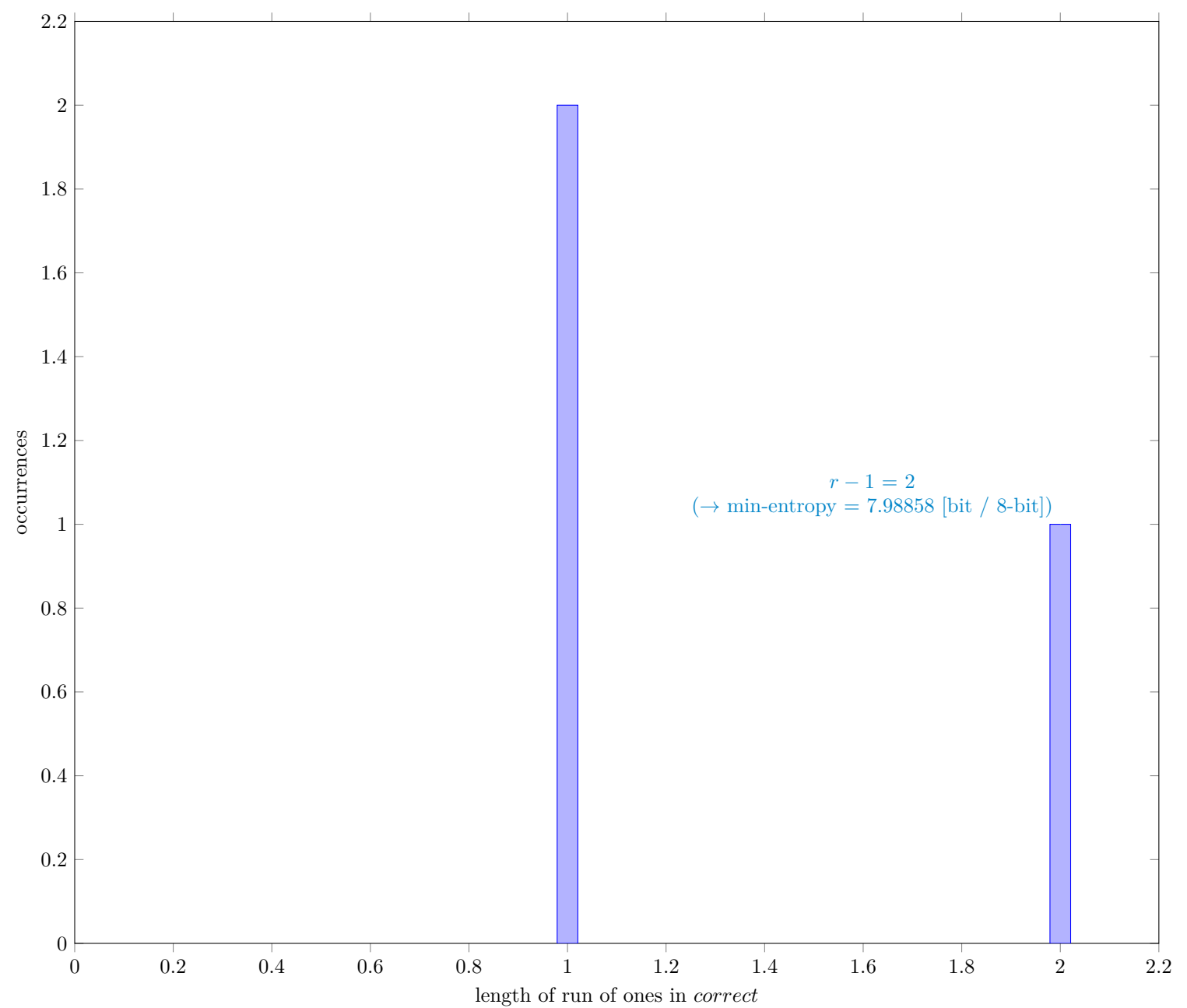


Fig. 8 Distribution of *correct*

3.4.1 Supplemental information for traceability

Table 8 Supplemental information for traceability (NIST SP 800-90B Section 6.3.7)

| Symbol               | Value      |
|----------------------|------------|
| $N$                  | 999937     |
| $C$                  | 3779       |
| $P_{\text{global}}$  | 0.00377924 |
| $P'_{\text{global}}$ | 0.00393729 |
| $r$                  | 3          |
| $P_{\text{local}}$   | 0.00215965 |

### 3.5 Lag Prediction Estimate (NIST SP 800-90B Section 6.3.8)

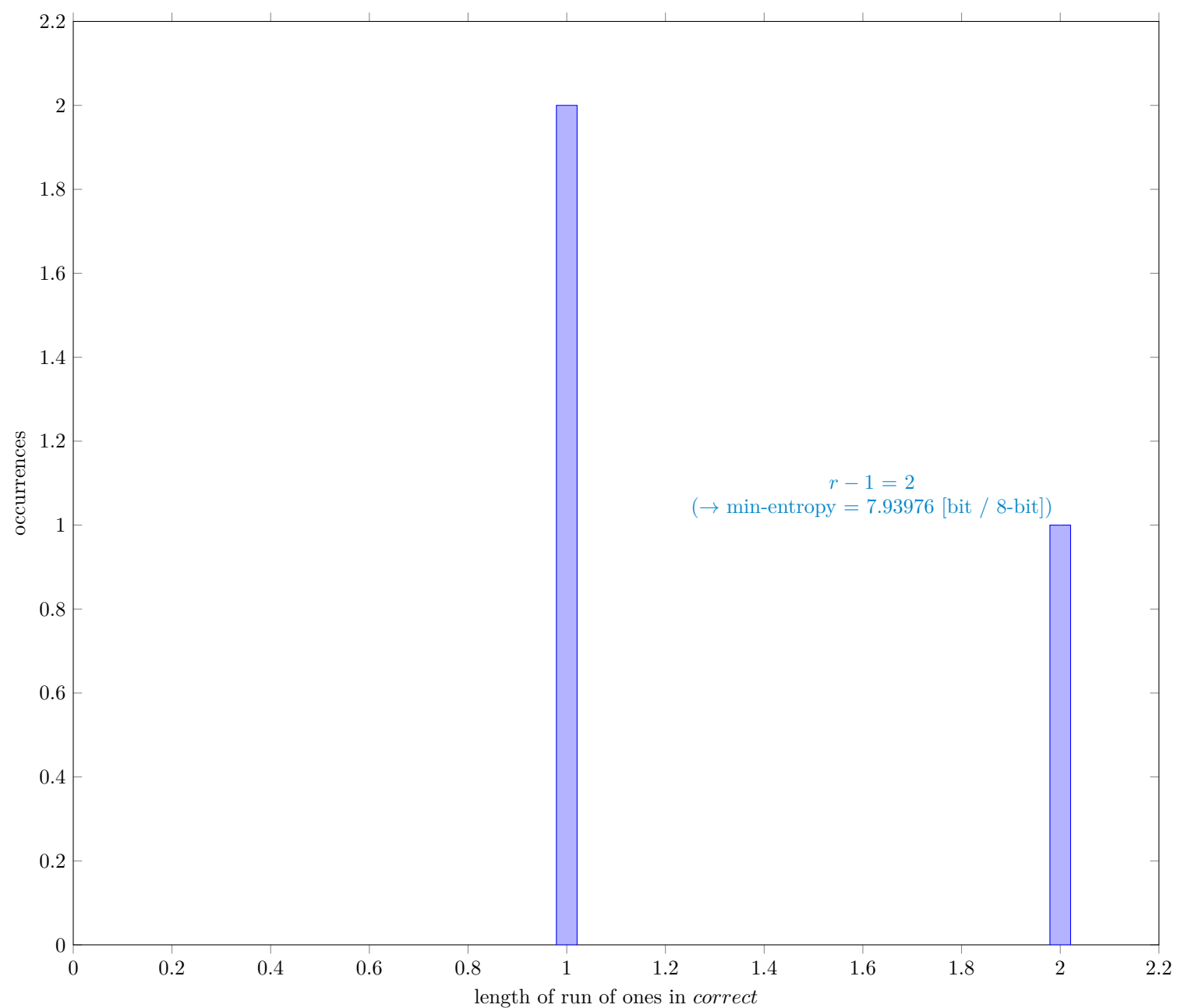


Fig. 9 Distribution of *correct*

#### 3.5.1 Supplemental information for traceability

Table 9 Supplemental information for traceability (NIST SP 800-90B Section 6.3.8)

| Symbol               | Value     |
|----------------------|-----------|
| $N$                  | 999999    |
| $C$                  | 3912      |
| $P_{\text{global}}$  | 0.003912  |
| $P'_{\text{global}}$ | 0.0040728 |
| $r$                  | 3         |
| $P_{\text{local}}$   | 0.0021596 |

### 3.6 The MultiMMC Prediction Estimate (NIST SP 800-90B Section 6.3.9)

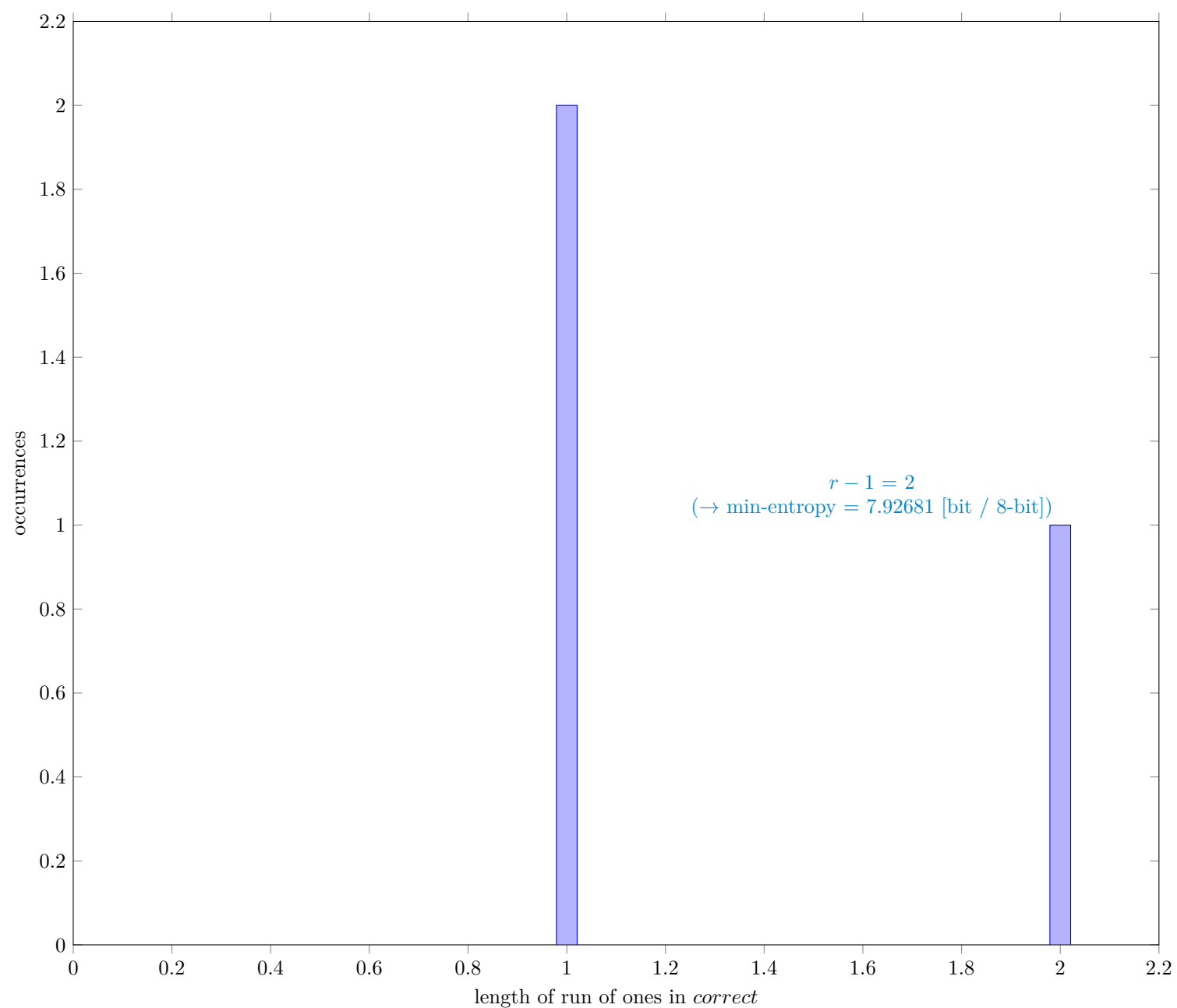


Fig. 10 Distribution of *correct*

#### 3.6.1 Supplemental information for traceability

Table 10 Supplemental information for traceability (NIST SP 800-90B Section 6.3.9)

| Symbol               | Value      |
|----------------------|------------|
| $N$                  | 999998     |
| $C$                  | 3948       |
| $P_{\text{global}}$  | 0.00394801 |
| $P'_{\text{global}}$ | 0.00410954 |
| $r$                  | 3          |
| $P_{\text{local}}$   | 0.0021596  |



3.7 The LZ78Y Prediction Estimate (NIST SP 800-90B Section 6.3.10)

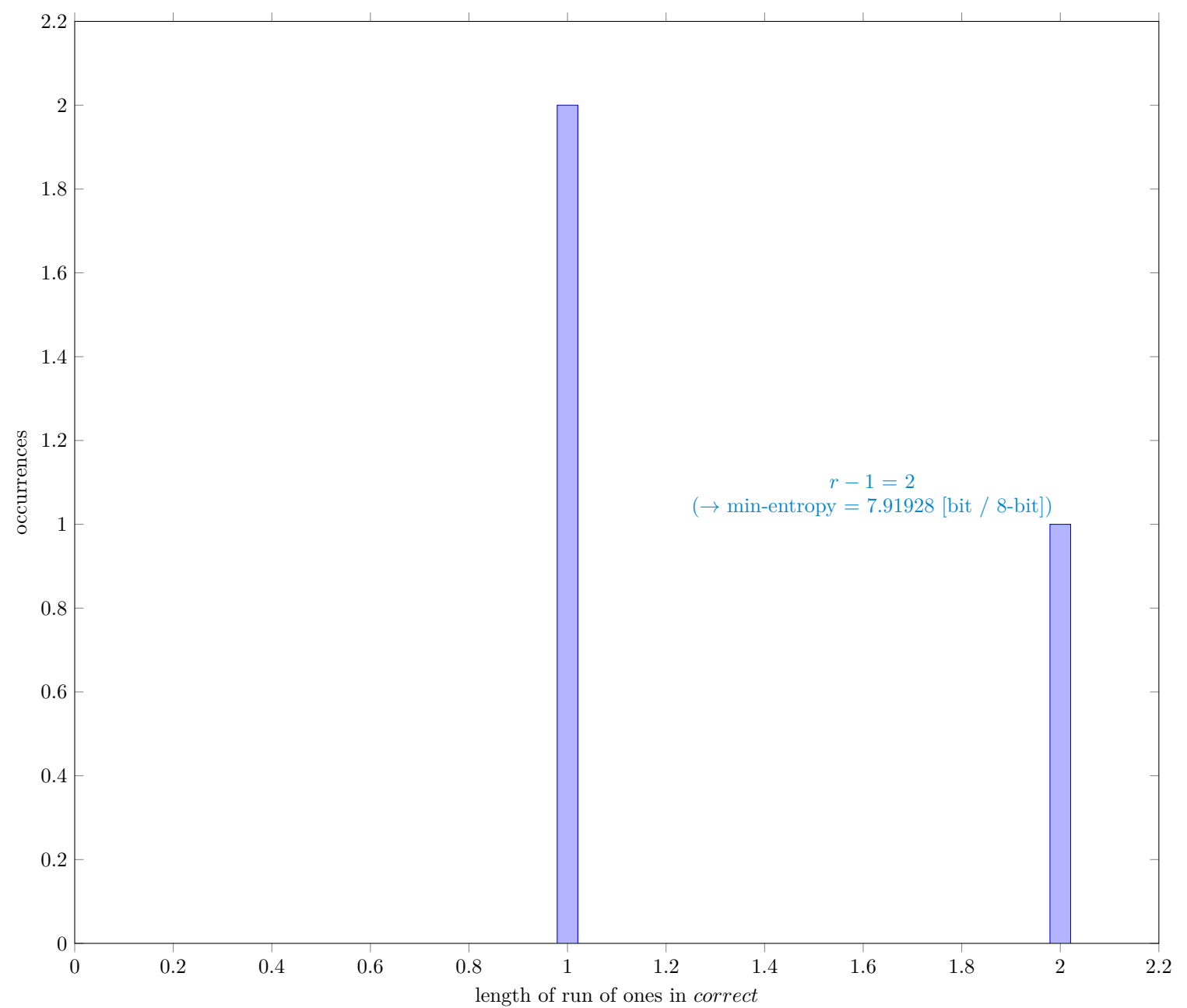


Fig. 11 Distribution of *correct*

3.7.1 Supplemental information for traceability

Table 11 Supplemental information for traceability (NIST SP 800-90B Section 6.3.10)

| Symbol               | Value      |
|----------------------|------------|
| $N$                  | 999983     |
| $C$                  | 3969       |
| $P_{\text{global}}$  | 0.00396907 |
| $P'_{\text{global}}$ | 0.00413103 |
| $r$                  | 3          |
| $P_{\text{local}}$   | 0.00215961 |

## 4 Detailed results of analysis by interpreting each sample as bitstrings

### 4.1 The Most Common Value Estimate (NIST SP 800-90B Section 6.3.1)

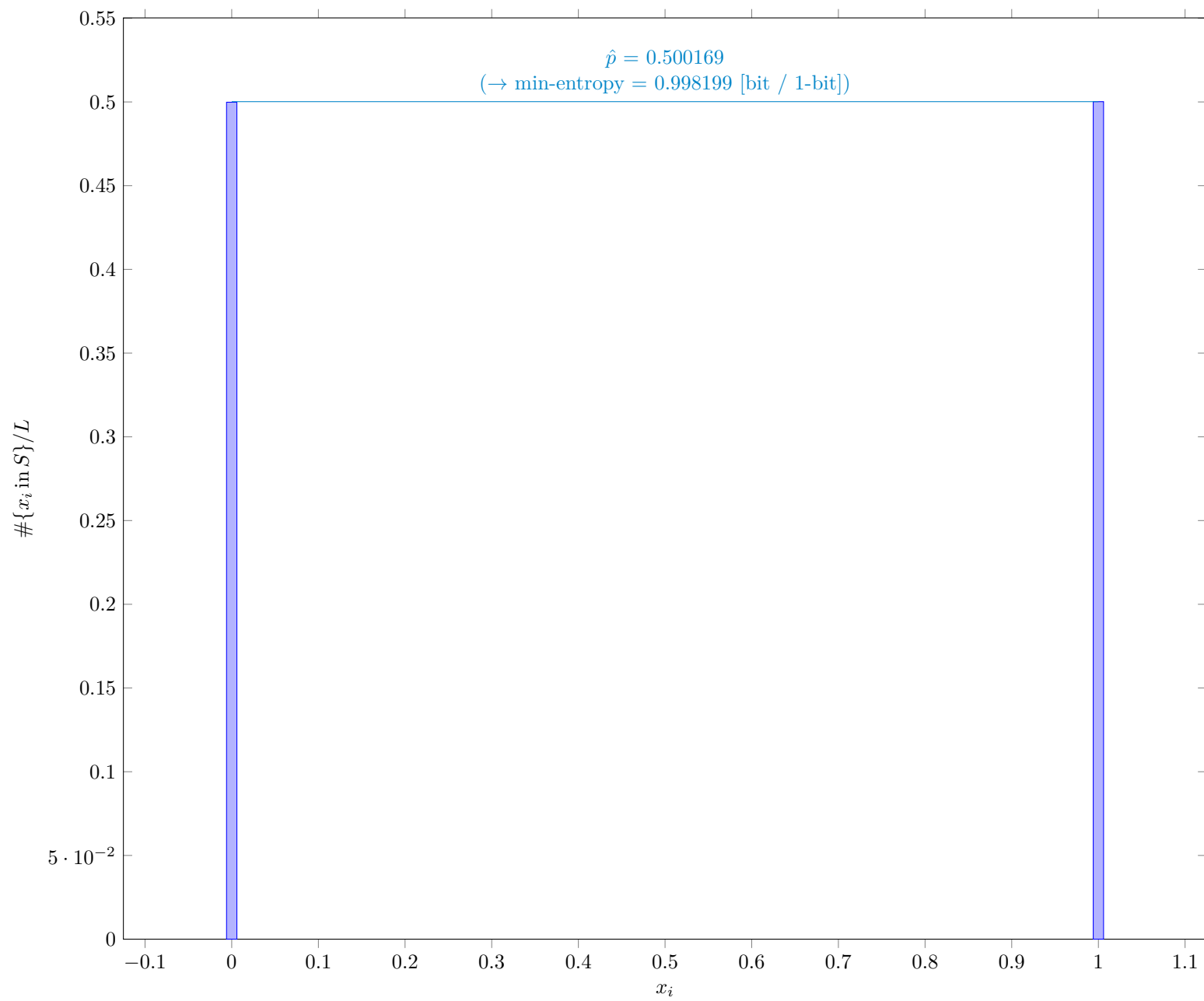


Fig. 12 Distribution of  $x_i$

#### 4.1.1 Supplemental information for traceability

Table 12 Supplemental information for traceability (NIST SP 800-90B Section 6.3.1)

| Symbol    | Value    |
|-----------|----------|
| mode      | 4001353  |
| $\hat{p}$ | 0.500169 |
| $p_u$     | 0.500624 |



## 4.2 The Collision Estimate (NIST SP 800-90B Section 6.3.2)

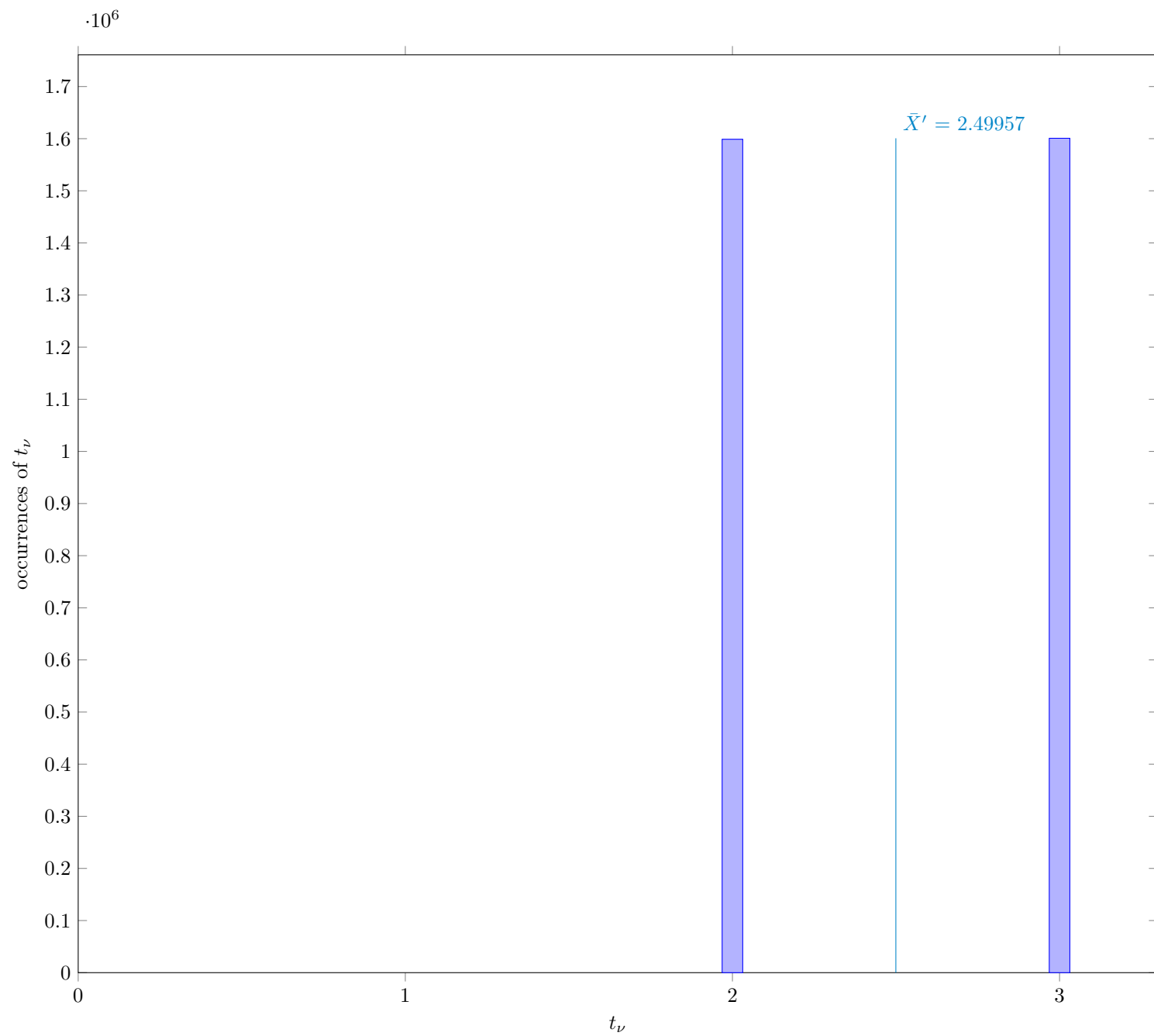


Fig. 13 Distribution of intermediate value  $t_\nu$

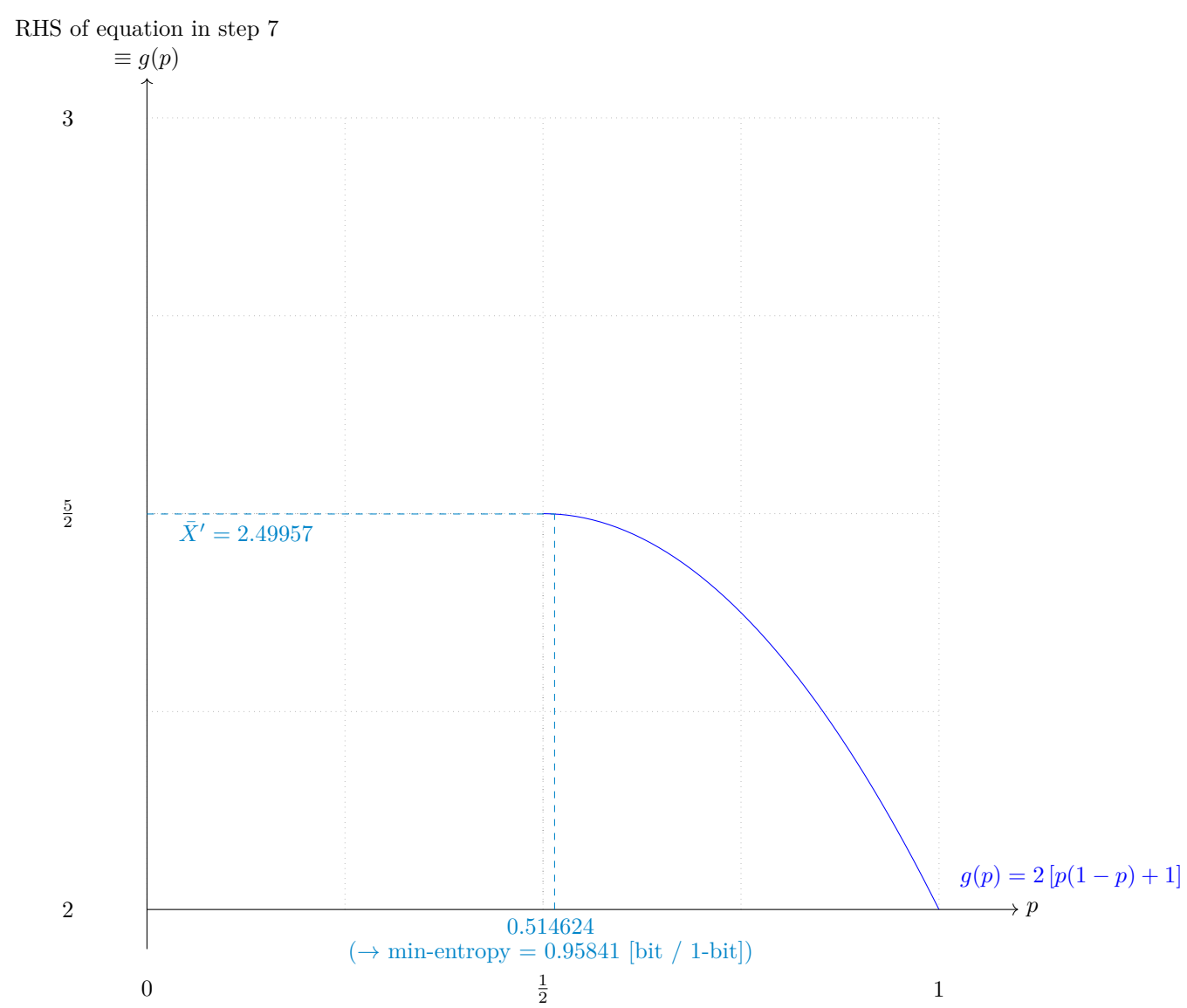


Fig. 14 Solution to the equation in step 7

4.2.1 Supplemental information for traceability

Table 13 Supplemental information for traceability (NIST SP 800-90B Section 6.3.2)

| Symbol         | Value    |
|----------------|----------|
| $p$            | 0.514624 |
| $\bar{X}$      | 2.50029  |
| $\bar{X}'$     | 2.49957  |
| $\hat{\sigma}$ | 0.5      |

### 4.3 The Markov Estimate (NIST SP 800-90B Section 6.3.3)

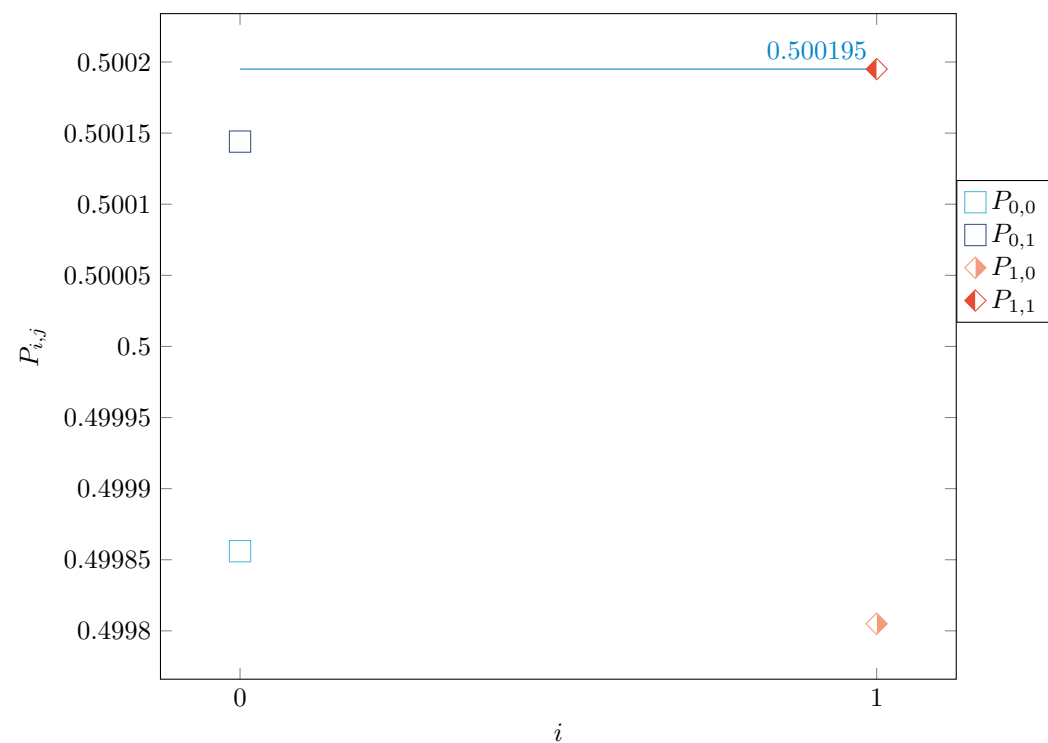


Fig. 15 Transition probability  $P_{i,j}$  of §6.3.3 of NIST SP 800-90B

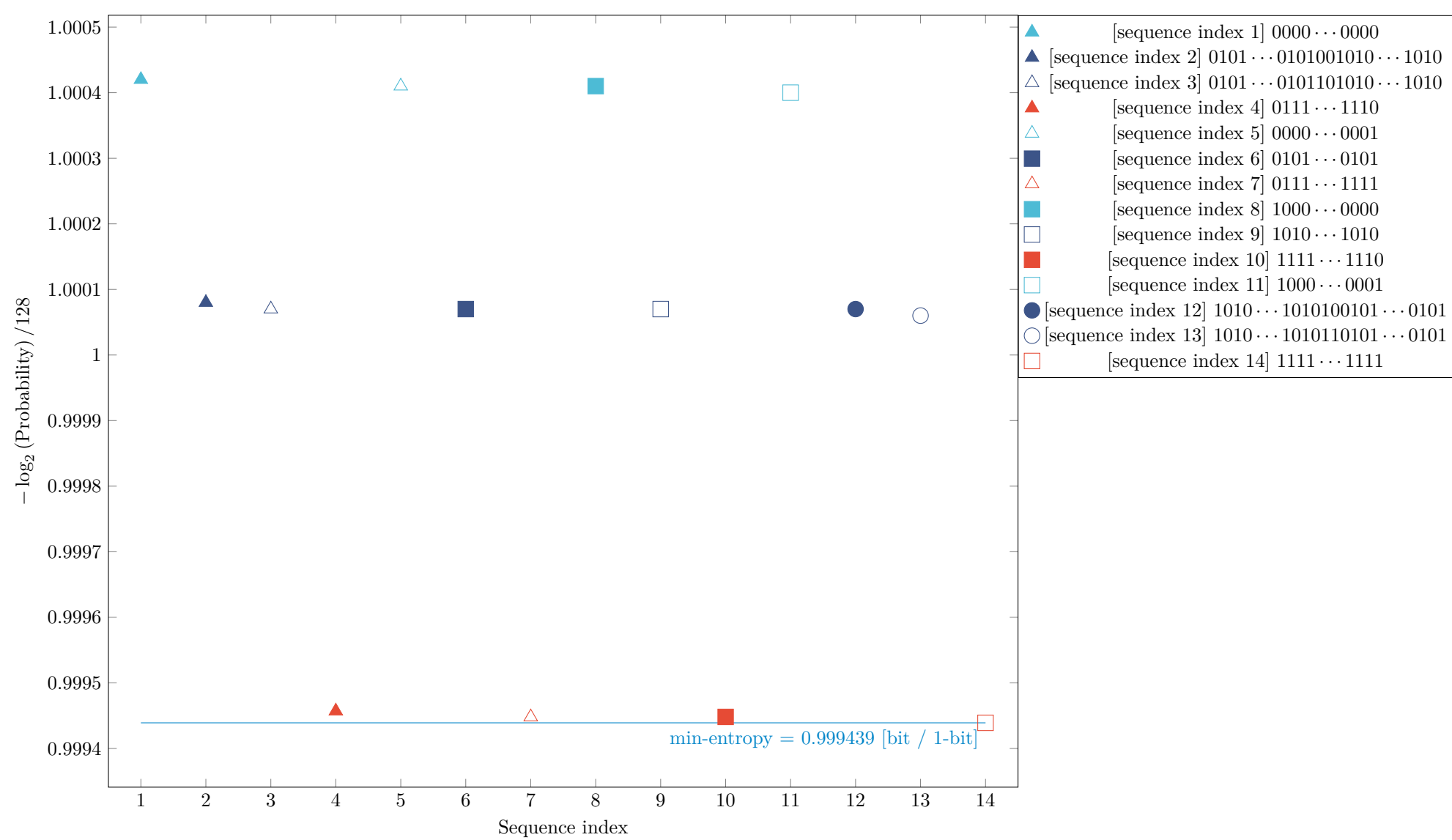


Fig. 16 Estimated Min-Entropy using §6.3.3 of NIST SP 800-90B

#### 4.4 The Compression Estimate (NIST SP 800-90B Section 6.3.4)

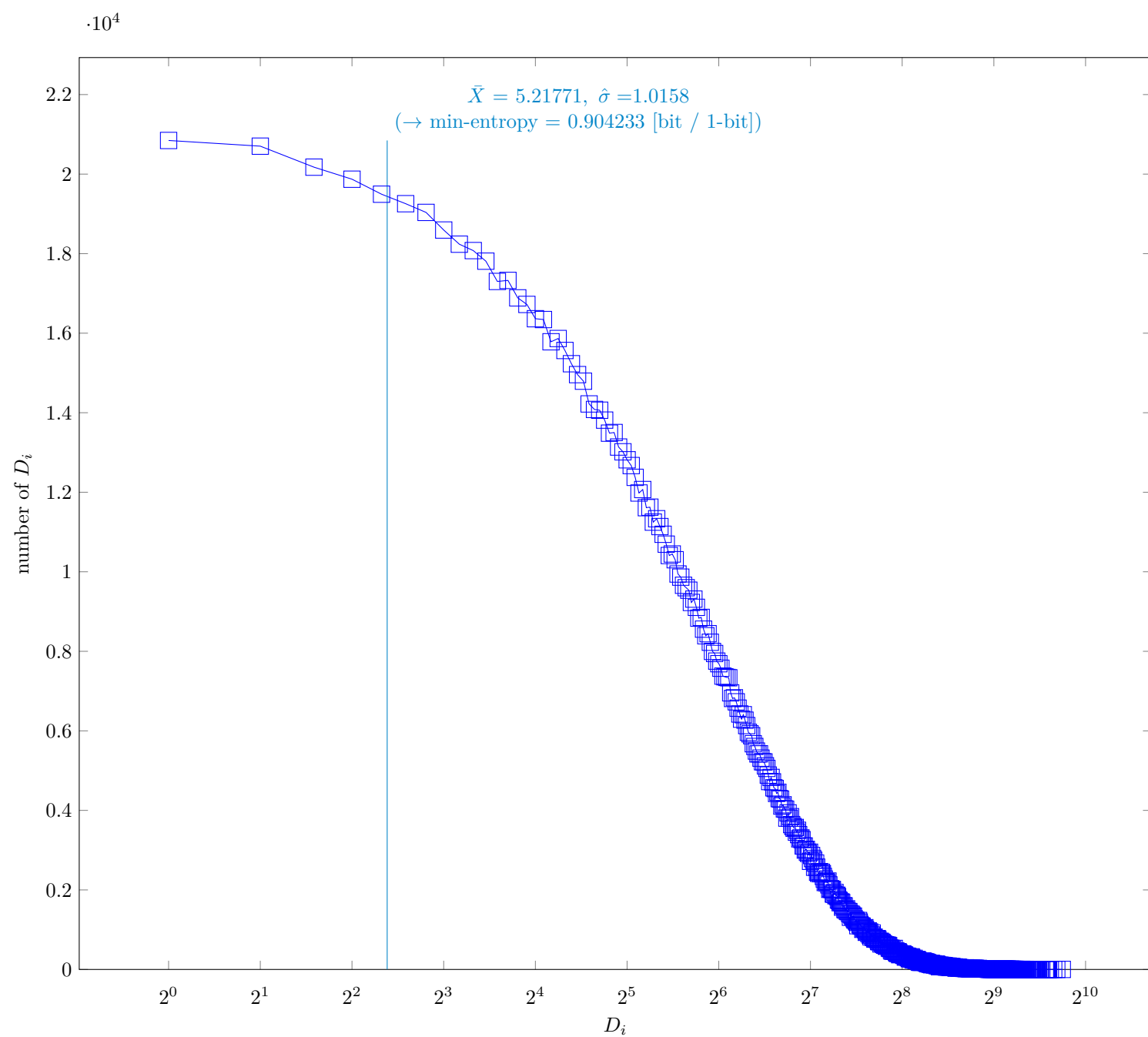


Fig. 17 Distribution of intermediate value  $D_i$

##### 4.4.1 Supplemental information for traceability

Table 14 Supplemental information for traceability (NIST SP 800-90B Section 6.3.4)

| Symbol         | Value     |
|----------------|-----------|
| $p$            | 0.0232698 |
| $\bar{X}$      | 5.21771   |
| $\hat{\sigma}$ | 1.0158    |
| $\bar{X}'$     | 5.21545   |

#### 4.5 The t-tuple Estimate (NIST SP 800-90B Section 6.3.5)

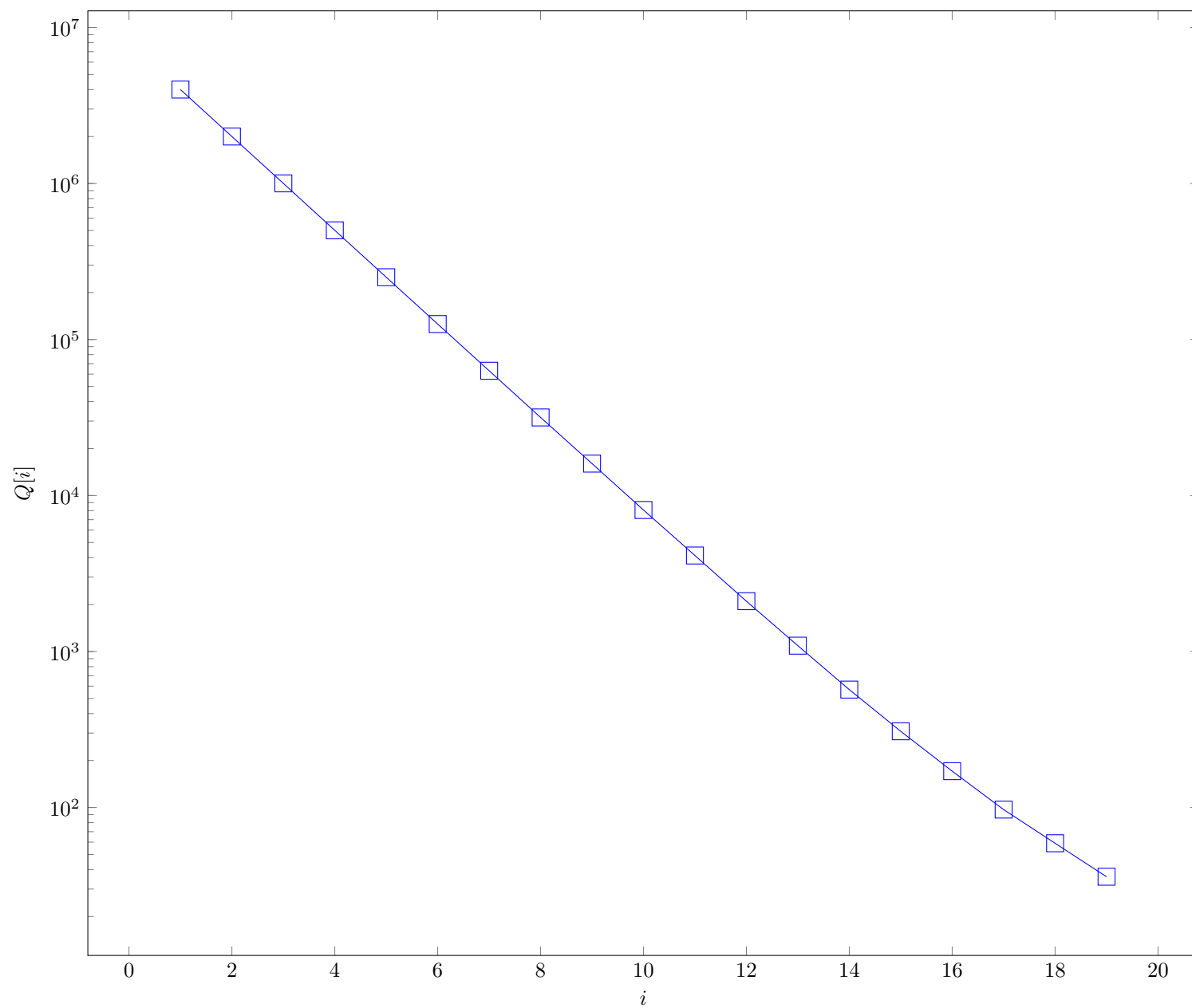


Fig. 18 Intermediate value  $Q[i]$  in §6.3.5 of NIST SP 800-90B



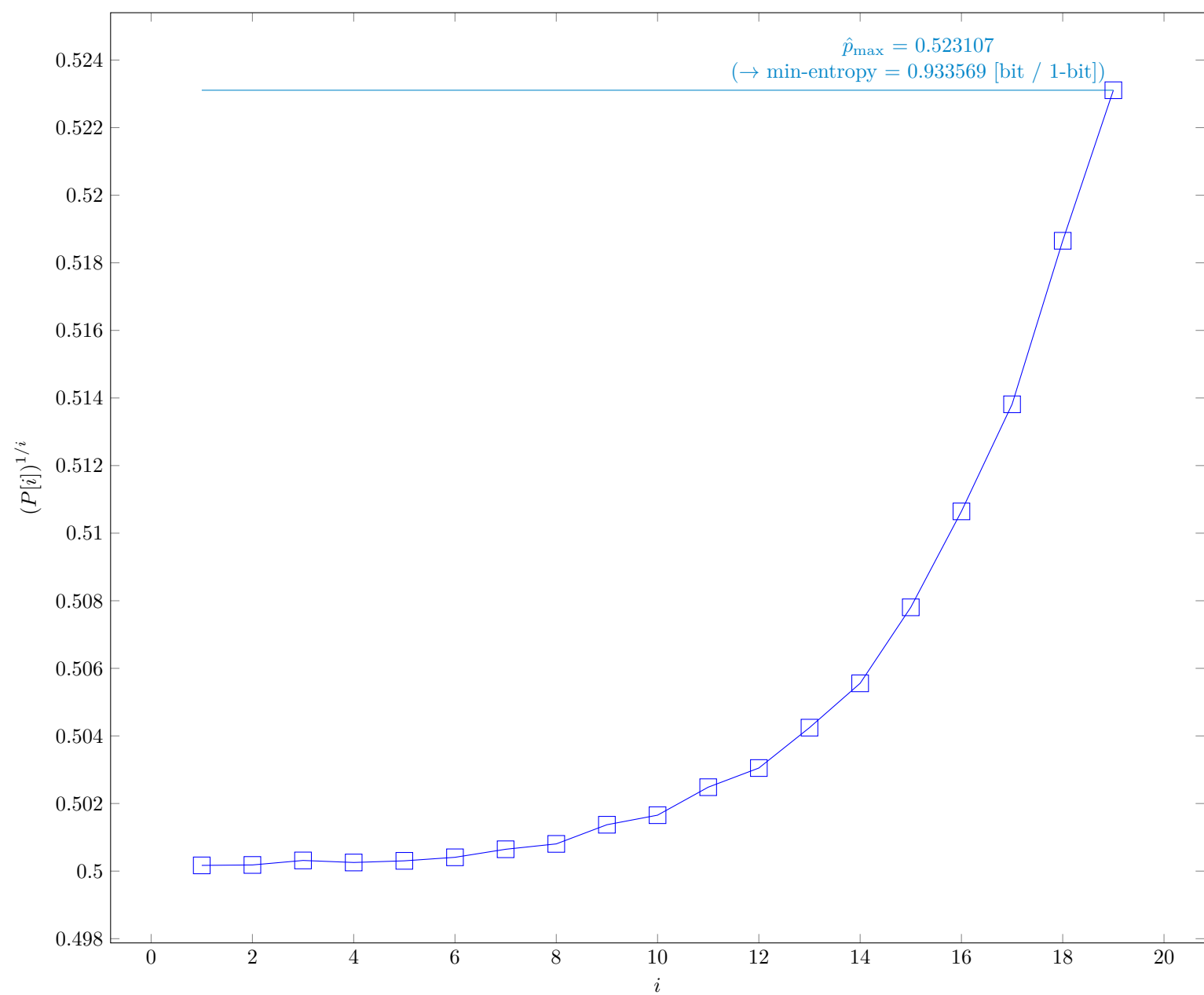


Fig. 19  $P[i]^{1/i}$  in §6.3.5 of NIST SP 800-90B

4.5.1 Supplemental information for traceability

Table 15 Supplemental information for traceability (NIST SP 800-90B Section 6.3.5)

| Symbol           | Value    |
|------------------|----------|
| $t$              | 19       |
| $\hat{p}_{\max}$ | 0.523107 |
| $p_u$            | 0.523561 |



#### 4.6 The LRS Estimate (NIST SP 800-90B Section 6.3.6)

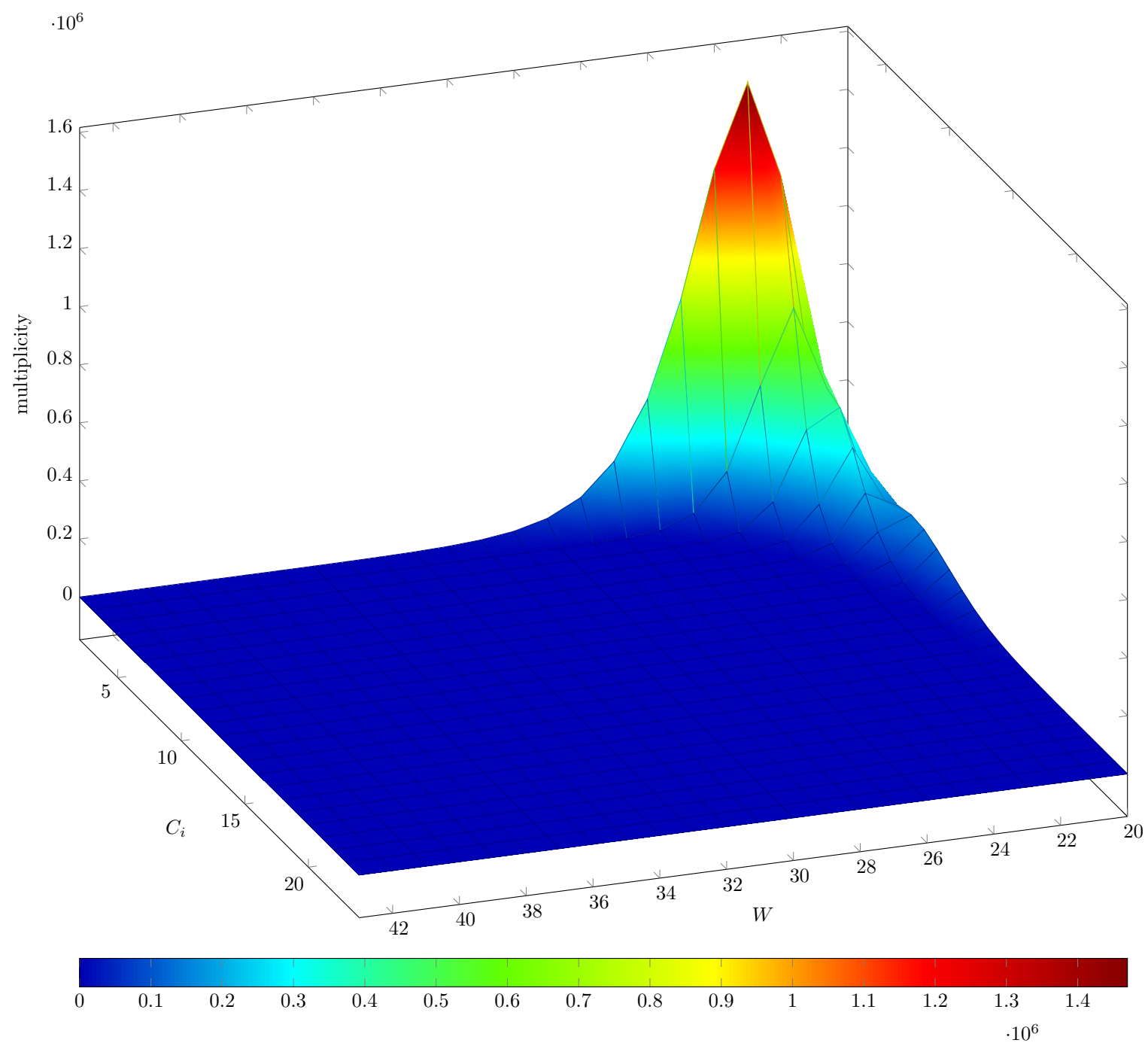


Fig. 20 Distribution of the multiplicity of  $W$ -tuple having the same  $C_i$  value in Step 3 of §6.3.6 of NIST SP 800-90B

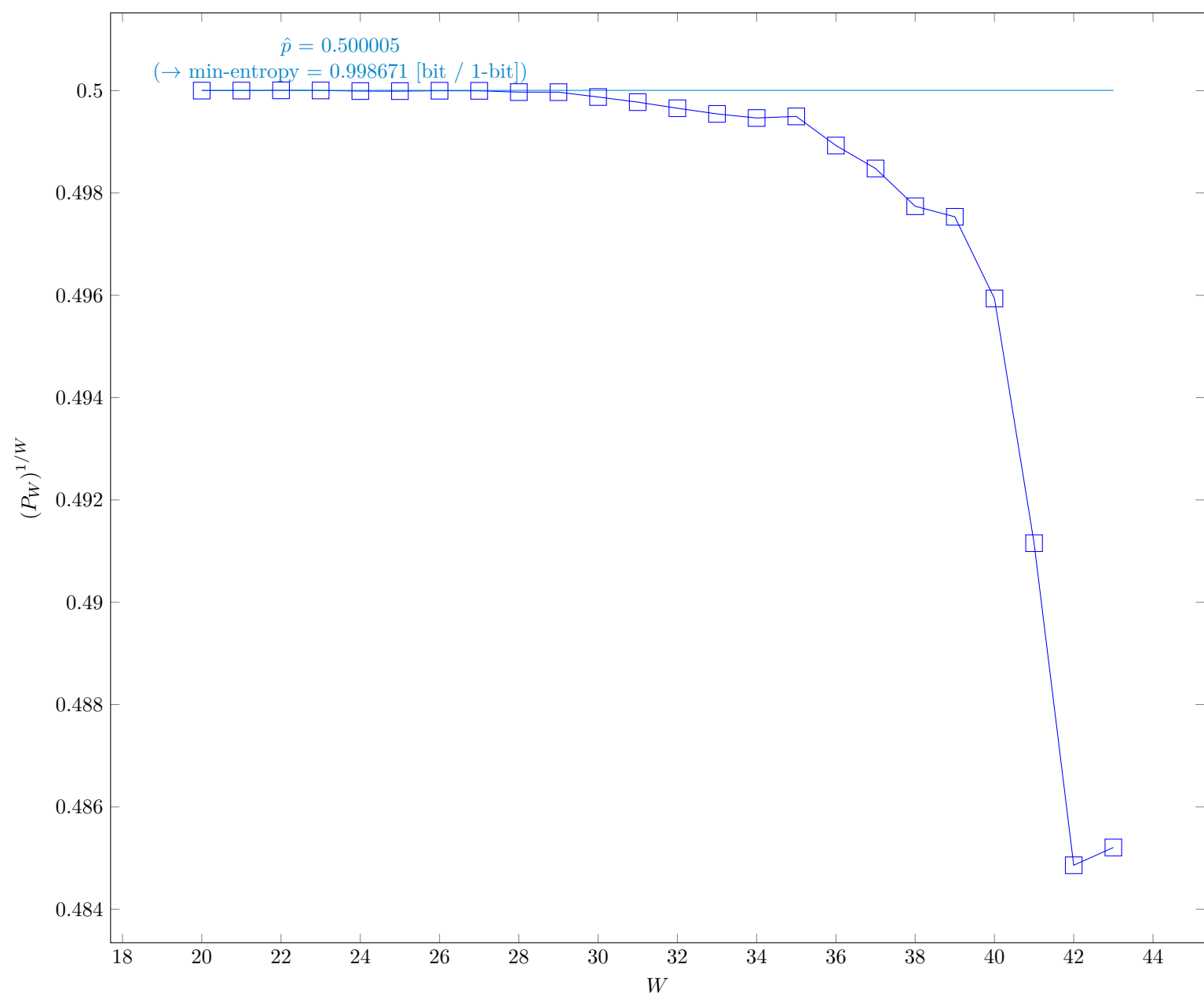


Fig. 21 Estimated average collision probability per string symbol in Step 3 of §6.3.6 of NIST SP 800-90B

4.6.1 Supplemental information for traceability

Table 16 Supplemental information for traceability (NIST SP 800-90B Section 6.3.6)

| Symbol    | Value    |
|-----------|----------|
| $u$       | 20       |
| $v$       | 43       |
| $\hat{p}$ | 0.500005 |
| $p_u$     | 0.500461 |

4.7 Multi Most Common in Window Prediction Estimate (NIST SP 800-90B Section 6.3.7)

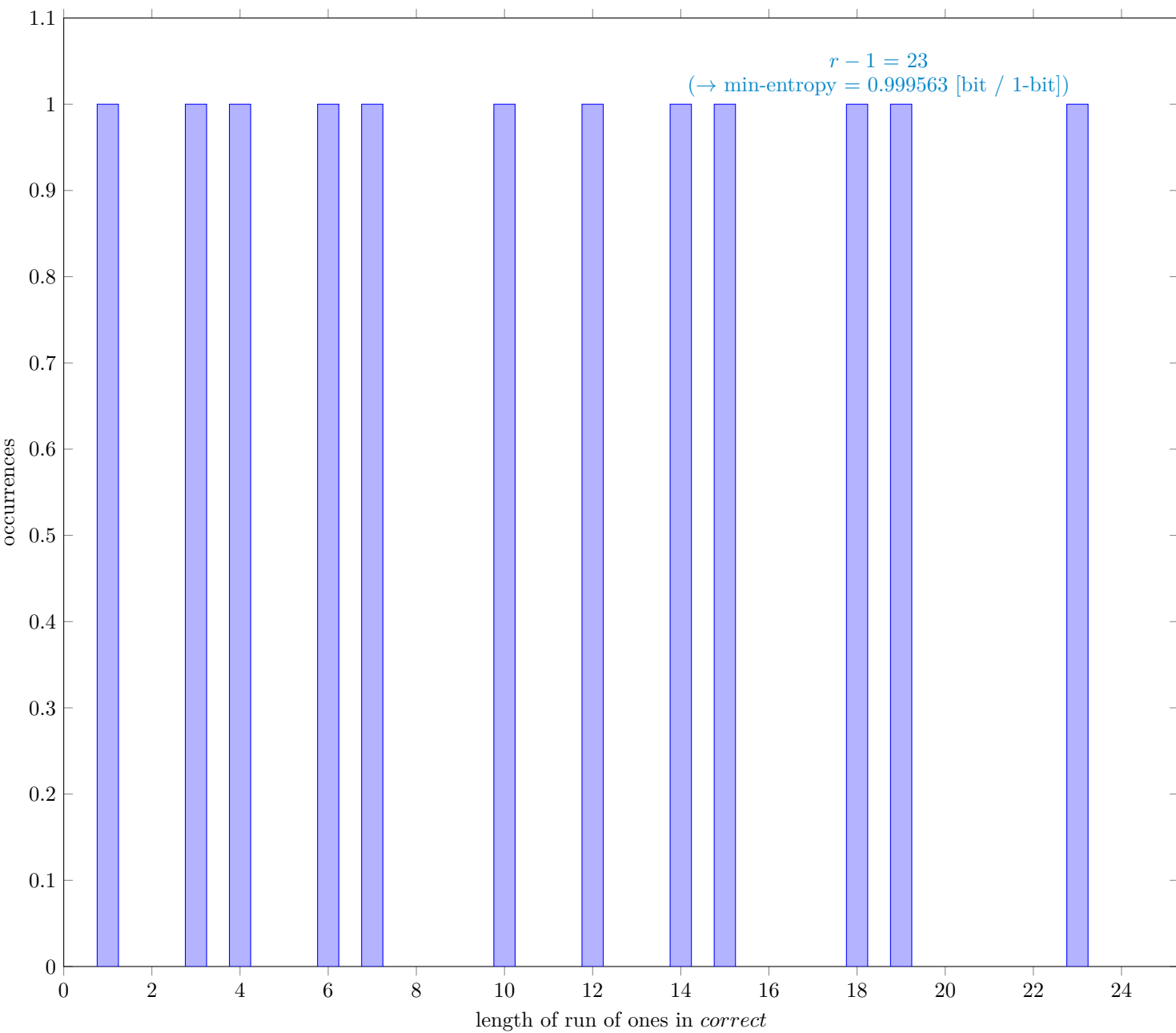


Fig. 22 Distribution of *correct*

4.7.1 Supplemental information for traceability

Table 17 Supplemental information for traceability (NIST SP 800-90B Section 6.3.7)

| Symbol               | Value    |
|----------------------|----------|
| $N$                  | 7999937  |
| $C$                  | 3997538  |
| $P_{\text{global}}$  | 0.499696 |
| $P'_{\text{global}}$ | 0.500152 |
| $r$                  | 24       |
| $P_{\text{local}}$   | 0.436006 |

#### 4.8 Lag Prediction Estimate (NIST SP 800-90B Section 6.3.8)

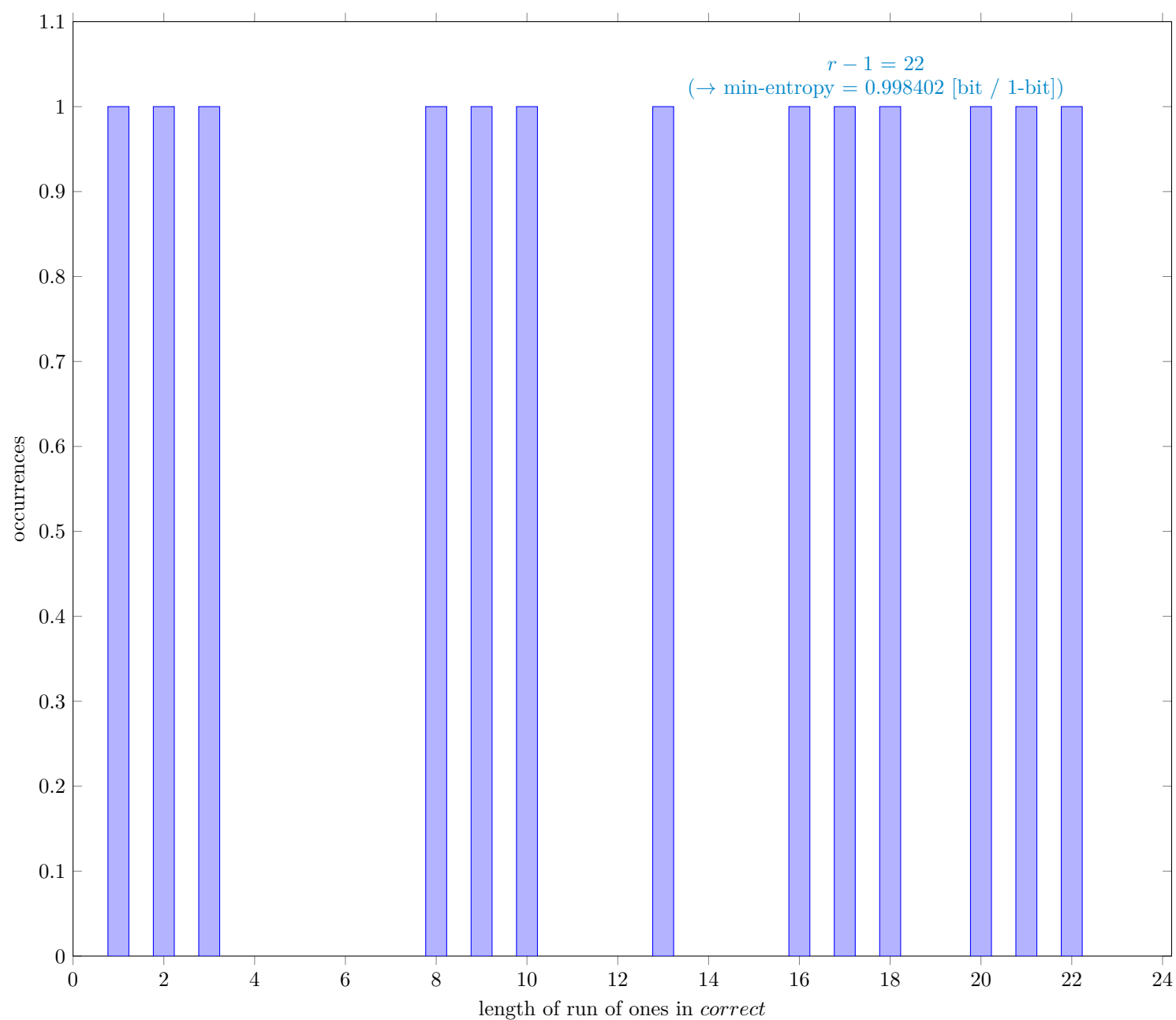


Fig. 23 Distribution of *correct*

##### 4.8.1 Supplemental information for traceability

Table 18 Supplemental information for traceability (NIST SP 800-90B Section 6.3.8)

| Symbol               | Value    |
|----------------------|----------|
| $N$                  | 7999999  |
| $C$                  | 4000791  |
| $P_{\text{global}}$  | 0.500099 |
| $P'_{\text{global}}$ | 0.500554 |
| $r$                  | 23       |
| $P_{\text{local}}$   | 0.42004  |

#### 4.9 The MultiMMC Prediction Estimate (NIST SP 800-90B Section 6.3.9)

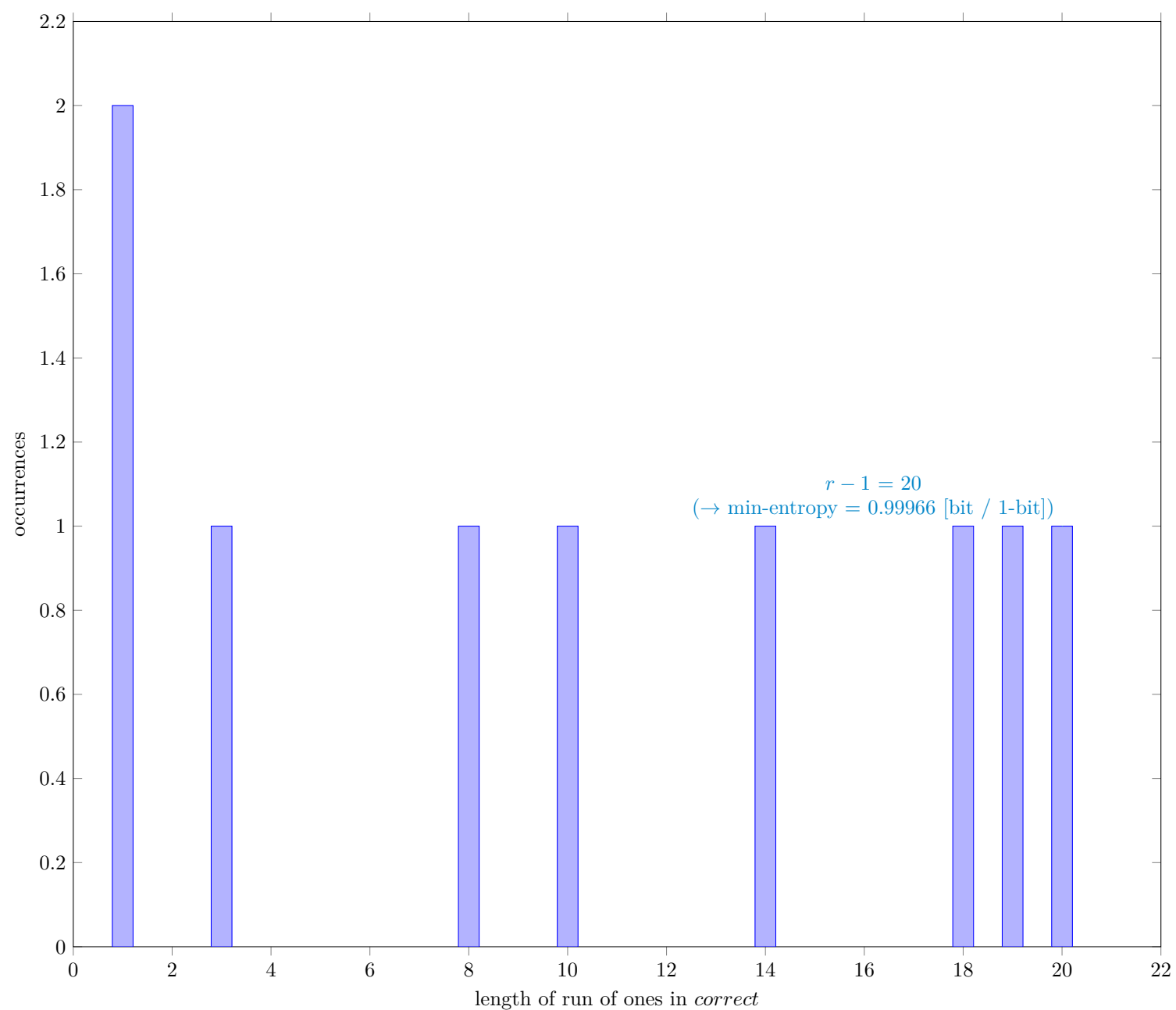


Fig. 24 Distribution of *correct*

##### 4.9.1 Supplemental information for traceability

Table 19 Supplemental information for traceability (NIST SP 800-90B Section 6.3.9)

| Symbol               | Value    |
|----------------------|----------|
| $N$                  | 7999998  |
| $C$                  | 3997298  |
| $P_{\text{global}}$  | 0.499662 |
| $P'_{\text{global}}$ | 0.500118 |
| $r$                  | 21       |
| $P_{\text{local}}$   | 0.385677 |



#### 4.10 The LZ78Y Prediction Estimate (NIST SP 800-90B Section 6.3.10)

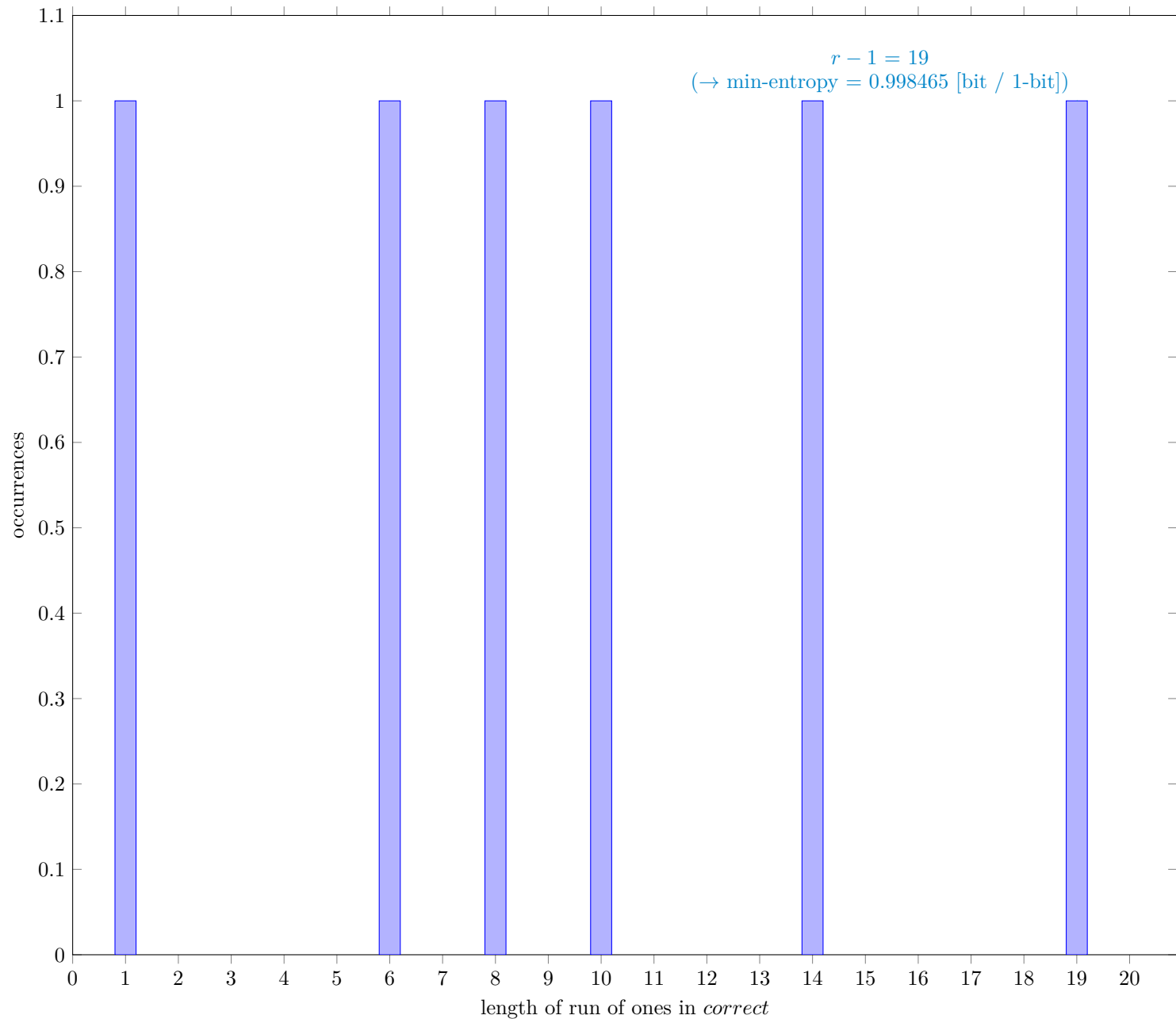


Fig. 25 Distribution of *correct*

##### 4.10.1 Supplemental information for traceability

Table 20 Supplemental information for traceability (NIST SP 800-90B Section 6.3.10)

| Symbol               | Value    |
|----------------------|----------|
| $N$                  | 7999983  |
| $C$                  | 4000606  |
| $P_{\text{global}}$  | 0.500077 |
| $P'_{\text{global}}$ | 0.500532 |
| $r$                  | 20       |
| $P_{\text{local}}$   | 0.36719  |

## 5 References

- [1] Meltem Sönmez Turan, Elaine Barker, John Kelsey, Kerry A. McKay, Mary L. Baish, Mike Boyle, *Recommendation for the Entropy Sources Used for Random Bit Generation*, NIST Special Publication 800-90B, Jan. 2018 <https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-90B.pdf>
- [2] G. Sakurai, *Proposed list of corrections for NIST SP 800-90B 6.3 Estimators*, June 2025 [https://github.com/g-g-sakura/AnotherEntropyEstimationTool/blob/main/documentation/ProposedListOfCorrections\\_SP800-90B.pdf](https://github.com/g-g-sakura/AnotherEntropyEstimationTool/blob/main/documentation/ProposedListOfCorrections_SP800-90B.pdf)
- [3] Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein, *Introduction to Algorithms (fourth edition)*, The MIT Press. <https://mitpress.mit.edu/9780262046305/introduction-to-algorithms/>
- [4] G. Sakurai, *Implementation Notes for entropy estimation based on NIST SP800-90B non-IID track*, Sep. 2025 [https://github.com/g-g-sakura/AnotherEntropyEstimationTool/blob/main/documentation/SP800-90B\\_EntropyEstimate\\_ImplementationNotes.pdf](https://github.com/g-g-sakura/AnotherEntropyEstimationTool/blob/main/documentation/SP800-90B_EntropyEstimate_ImplementationNotes.pdf)